



**Grade 5
General
Mathematics**



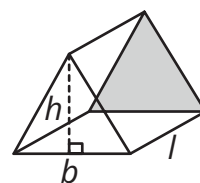
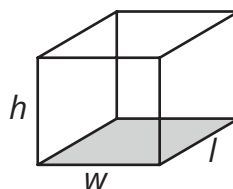
NSCAS NEBRASKA STUDENT-CENTERED ASSESSMENT SYSTEM

Mathematics Reference Sheet

2-Dimensional Shape	Area	Circumference
Circle	$A = \pi r^2$	$C = \pi d = 2\pi r$
Triangle	$A = \frac{1}{2}bh$	Perimeter
Rectangle	$A = l \times w$	$P = 2l + 2w = 2(l + w)$
Square	$A = s \times s$	$P = s + s + s + s$
Trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$	
Parallelogram	$A = bh$	

3-Dimensional Shape	Volume	Surface Area
Rectangular Prism	$V = lwh = Bh$	SA = the sum of the areas of all the faces
Triangular Prism	$V = \frac{1}{2}bhl = Bl$	SA = the sum of the areas of all the faces
Cone	$V = \frac{1}{3}\pi r^2h$	
Cylinder	$V = \pi r^2h$	
Sphere	$V = \frac{4}{3}\pi r^3$	

In the images to the right, the shaded faces are the bases.



2-Dimensional Shape Key		
b = base length	l = length	s = side
h = height	w = width	
d = diameter	r = radius	

Use 3.14 for π .

3-Dimensional Shape Key		
d = diameter	l = length	w = width
r = radius	h = height of shape	B = area of base
	b = base length	

Percent Change	Pythagorean Theorem
% change = $\frac{\text{difference in amounts}}{\text{original amount}}$	$c^2 = a^2 + b^2$

Standard Units	Metric Units
Conversions – Length	
1 foot (ft) = 12 inches (in.)	1 centimeter (cm) = 10 millimeters (mm)
1 yard (yd) = 3 feet (ft) = 36 inches (in.)	1 meter (m) = 100 centimeters (cm)
1 mile (mi) = 1,760 yards (yd) = 5,280 feet (ft)	1 meter (m) = 1,000 millimeters (mm)
	1 kilometer (km) = 1,000 meters (m)
Conversions – Volume	
1 cup = 8 fluid ounces (fl oz)	1 liter (l) = 1,000 milliliters (ml)
1 pint (pt) = 2 cups	1 liter (l) = 1,000 cubic centimeters (cu cm)
1 quart (qt) = 2 pints (pt)	
1 gallon (gal) = 4 quarts (qt)	
Conversions – Weight/Mass	
1 pound (lb) = 16 ounces (oz)	1 gram (g) = 1,000 milligrams (mg)
1 ton = 2,000 pounds (lb)	1 kilogram (kg) = 1,000 grams (g)

Directions:

On the following pages of your practice test booklet are questions for the Grade 5 *Nebraska Student-Centered Assessment System (NSCAS) General Mathematics* practice test.

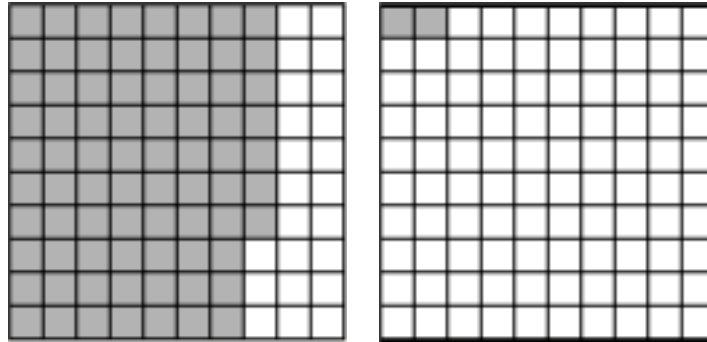
Read these directions carefully before beginning the test.

This practice test will include several different types of questions. Multiple choice questions will ask you to select an answer from among four choices. Multiple select questions will ask you to select multiple correct answers from among five or more answer choices. For some questions, there may be two parts, Part A and Part B, where each part has a multiple choice or multiple select question. Some questions will ask you to construct an answer by following the directions given.

For all questions:

- Read each question carefully and choose the best answer.
- You may use scratch paper to solve the problems.
- The Mathematics Reference Sheet is provided in the back of the test booklet. You may refer to this page at any time during the test.
- Be sure to answer ALL the questions.

1. This question has two parts. Answer Part A, and then answer Part B.
Use the models to answer the question.



Part A

What is the sum of $0.77 + 0.02$?

- A. 0.772
- B. 0.79
- C. 0.792
- D. 0.97

Part B

What is the difference of $0.77 - 0.02$?

- A. 0.768
- B. 0.758
- C. 0.75
- D. 0.57

2. What is the product of 487×39 ?

- A. 703
- B. 5,844
- C. 16,033
- D. 18,993

3. A basketball team has played nine games so far this season. The list shows the number of points the team scored at each game.

58, 43, 54, 43, 55, 57, 39, 76, 82

Which prediction about the team's score in the next game can MOST LIKELY be made using the data?

- A. The team will likely score 43 points in the next game.
- B. The team will likely score more than 80 points in the next game.
- C. The team will likely score between 40 and 60 points in the next game.
- D. The team will likely score either less than 39 points or more than 82 points in the next game.

4. **This question has two parts. Answer Part A, and then answer Part B.**

A student measured the heights of some plants. This list shows the heights, in inches.



Part A

What is the difference between the heights of the middle two plants?

- A. $\frac{5}{8}$ inch B. $3\frac{5}{8}$ inches
C. $6\frac{3}{4}$ inches D. $6\frac{3}{8}$ inches

Part B

What is the difference between the heights of the tallest plant and the shortest plant?

- A. $1\frac{7}{16}$ inches B. $1\frac{7}{8}$ inches
C. $2\frac{1}{8}$ inches D. $2\frac{2}{3}$ inches

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5. A sign has a width of 254 millimeters.

What is the width of the sign in centimeters? Write the answer in the box.

centimeters

6. What is $[(16 - 6) \div 2] + (3 \times 3)$?

- A. 14
- B. 22
- C. 24
- D. 48

7. Which values make the comparison true? Write a value in each box.

$7\frac{1}{2}$

$7\frac{1}{4}$

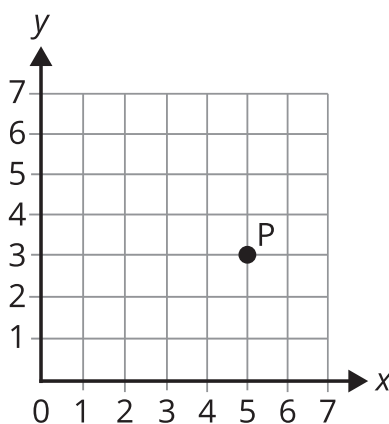
$7\frac{25}{100}$

7.04

7.250

$< 7.25 <$

8. Use the coordinate grid below to answer the question.



What are the coordinates of point P?

- A. (3, 5)
 - B. (5, 3)
 - C. (5, 4)
 - D. (6, 3)
9. What is $4,376 \div 36$?
- A. 121
 - B. 121 R20
 - C. 122
 - D. 122 R16

10. Which list orders the numbers from least to greatest?

- A. 6.25, 6.5, $6\frac{4}{5}$, $6\frac{3}{4}$
- B. 6.5, 6.25, $6\frac{3}{4}$, $6\frac{4}{5}$
- C. 6.25, 6.5, $6\frac{3}{4}$, $6\frac{4}{5}$
- D. 6.5, $6\frac{4}{5}$, 6.25, $6\frac{3}{4}$

11. Use the figure to answer the question.



Which words or phrases BEST describe the figure?

Select **three**.

- A. square
- B. rectangle
- C. trapezoid
- D. The figure has exactly four angles.
- E. The figure has four sides of equal length.
- F. The figure has exactly one set of parallel sides.
- G. The figure has exactly two sets of parallel sides.

12. What is 321×254 ?

A. 67,089

B. 71,434

C. 80,534

D. 81,534

13. What is the standard form of $90,000 + 800 + 50 + 2 + 0.1 + 0.007$?

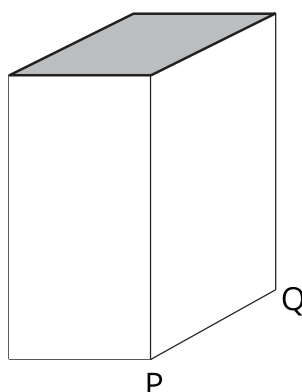
A. 90,852.107

B. 90,852.017

C. 98,052.107

D. 98,052.017

14. Use the rectangular prism to complete the statements.



Select the correct choices to complete the sentences about a rectangular prism.

Line segment PQ is called a(n) (**edge, face, vertex**).

The shaded part of the rectangular prism is called a(n) (**edge, face, vertex**).

The number of edges of the rectangular prism is (**6 8 12**).

15. On Sunday, a student starts reading a book with 140 pages. Each day, the student records the page number he is on when he finishes reading for that day. The table shows this data.

Daily Reading Log

Day	Ending Page Number
Sunday	21
Monday	41
Tuesday	60
Wednesday	79
Thursday	
Friday	
Saturday	

Which **two** predictions can be made using the data in the table?

- A. The student will read about 10 pages on Friday.
 - B. The student will finish reading the book on Friday.
 - C. The student will read about 20 pages on Thursday.
 - D. The student will finish reading the book on Saturday.
 - E. The student will start reading a new book on Saturday.
16. Evaluate.

$$32 \div 4 \times (20 - 16)$$

- A. 2
- B. 16
- C. 32
- D. 144

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17. What is the difference of $\frac{4}{5} - \frac{1}{3}$? Choose from the numbers given and write the answer in the blank.

1

2

3

5

7

8

9

15

$$\frac{4}{5} - \frac{1}{3} = \underline{\hspace{2cm}}$$

18. Use the table to answer the question.

Fourth Graders' Favorite Beverage

Beverage	Votes
Apple juice	
Orange juice	
Soda	
Milk	

The table shows the favorite beverage of the fifth-grade students. Based on the table, how many more students chose apple juice or orange juice than soda?

- A. 4
- B. 6
- C. 10
- D. 12

19. Use the rule to answer the question.

$$y = 4x$$

Which points are formed by the rule?

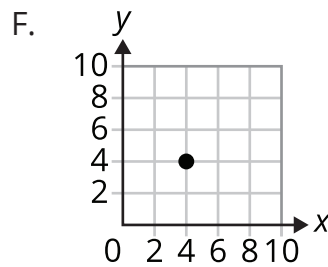
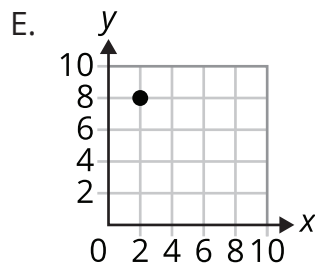
Select **three** options.

A. (0, 0)

B. (3, 4)

C. (8, 2)

D. (5, 20)



20. What is $\frac{1}{4} \div 3$?

A. $\frac{1}{12}$

B. $\frac{3}{4}$

C. $\frac{4}{3}$

D. 12

**NSCAS General Grade 5
Practice Test Answer Key
Mathematics (English)**



Sequence	Key	Points
1	Part A: B Part B: C	2
	Part A or Part B	1
2	D	1
3	C	1
4	Part A: A Part B: B	2
	Part A or Part B	1
5	25.4 or equivalent	1
6	A	1
7	$7.04 < 7.25 < 7\frac{1}{2}$	1
8	B	1
9	B	1
10	C	1
11	C, D, F	2
	2 of 3 correct responses	1
12	D	1
13	A	1
14	edge face 12	2
	2 of 3 correct responses	1
15	C, D	1
16	C	1
17	$\frac{4}{5} - \frac{1}{3} = \frac{7}{15}$	1
18	A	1
19	A, D, E	2
	2 of 3 correct responses	1
20	A	1